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CSST 104 (MACHINE LEARNING)
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From Data to Prediction: Machine Learning in Image Recognition Systems

Part A – Identify the ML Problem

Question	Your Answer
1. What is the application?	Image Recognition (e.g., identifying objects, faces, or animals in photos)
2. What problem are you trying to solve?	Automatically identifying what object, person, or thing is shown in a given photo or image
3. What are the inputs?	Digital images (pixel data), often labeled with what they contain (e.g., "cat," "dog," "car")
4. What is the expected output?	A label or category that correctly identifies what is in the image
5. What patterns or relationships might the ML system learn?	Certain arrangements of pixels, shapes, colors, and edges consistently correspond to specific objects (e.g., pointed ears and whiskers usually mean "cat")
6. What data would the system need?	A large collection of labeled images, meaning many photos where the correct object/category is already known and tagged

Part B – Traditional Programming vs. Machine Learning

Traditional Programming

Input → Algorithm → Output

If we used traditional programming, a developer would need to manually write rules describing exactly what makes an image a "cat" or a "car," such as specific pixel colors, shapes, or edge patterns. This is extremely difficult because images vary endlessly in lighting, angle, size, and background. It would be nearly impossible to write enough rules to cover every possible way a cat or car could appear in a photo.

Machine Learning

Input + Output/Data → Machine Learning Algorithm

Instead of hardcoding visual rules, we give the ML algorithm thousands of labeled images (input: the image, output: the correct label). The algorithm studies these examples and learns on its own which visual patterns are associated with each label. Once trained, it can look at a brand new, unlabeled image and correctly predict what it shows, without a human ever writing specific rules for shapes, colors, or edges.

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Part C – Create Your Own Mini Dataset

Dataset (at least 5 observations):

Input (Image contains: pointed ears, whiskers, fur)	Output (Label)
Yes, Yes, Yes	Cat
Yes, Yes, No	Cat
No, No, No	Not a Cat
No, No, Yes	Not a Cat
Yes, No, Yes	?

1. **Identify the pattern:** Images that show pointed ears and whiskers are consistently labeled "Cat," even if fur presence varies. Images missing both pointed ears and whiskers are labeled "Not a Cat."
2. **Determine the missing output:** For "Yes, No, Yes" (pointed ears present, no whiskers, fur present), the closest match to the pattern is still **Cat**, since pointed ears appear to be the strongest indicator in the data so far.
3. **Mathematical relationship/function:** This isn't a numeric formula since the data is categorical, but it can be written as a rule: *If pointed ears = Yes, then Label = Cat; otherwise, Label = Not a Cat.*
4. **How an ML algorithm could learn this:** By examining the paired examples, the algorithm looks for which input features most consistently predict the correct label. It would notice that "pointed ears = Yes" appears every time the label is "Cat," and adjust itself to treat that feature as the strongest signal. Once trained, it can apply this learned rule to a brand new image it has never seen and predict whether it shows a cat.

Part D – Explain the Training Process

How does a Machine Learning algorithm learn from data?

A machine learning algorithm learns by being given many example images, each paired with the correct label describing what the image contains. During training, the algorithm makes a prediction for each image, compares it to the actual correct label, and adjusts itself to reduce the mistakes it makes. Over many repetitions of this process, it gradually learns which visual patterns, such as shapes, colors, and textures, are associated with each category. Once training is complete, the algorithm can look at a completely new image it has never seen before and predict what it shows based on the patterns it learned. This is different from traditional programming because no one wrote direct rules for recognizing each object; the relationship was learned entirely from the data.

Reflection

Why is learning from data useful instead of manually programming every possible rule?

Learning from data is especially useful for image recognition because it would be impossible for a programmer to manually write rules covering every possible angle, lighting condition, size, and background in which an object could appear in a photo. Images are far too complex and varied for fixed rules to handle reliably. By allowing the machine to learn directly from thousands of labeled examples, the system can

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automatically pick up on subtle visual patterns that humans might not even think to program, making it far more accurate, flexible, and practical than trying to hand-code every rule.